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START

Web Scraping and Data Collection

Day 3: From Shared Workflow to Participants' Own Data

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Day 3: From Shared Workflow to Participants' Own Data

Exploration Checks Plausibility Before Any Inferential or Comparative Analysis

Exploratory analysis of web-derived data serves a different purpose than exploratory analysis of survey or experimental data. Its primary goal is **plausibility checking** rather than hypothesis generation. Key questions include:

- Do the row counts match expectations?
- Are distributions of key variables plausible given domain knowledge?
- Are there unexpected spikes, gaps, or outliers that may reflect extraction artifacts?
- Do missing values cluster in ways that suggest systematic extraction problems?

Descriptive Statistics Reveal Whether Cleaned Data Are Internally Consistent

Summary statistics for web-derived data should be interpreted with the following questions in mind:

- **Mean and median**: are they plausible given the domain?
- **Min and max**: do extreme values reflect real entities or extraction artifacts?
- **Standard deviation**: is the spread consistent with the known heterogeneity of the population?
- **Missing value count**: does it match what was expected from the cleaning log?

For the World Bank population dataset (Application 5), the maximum value of approximately 7.85 billion corresponds to the global total — a structural feature of the dataset, not an outlier.

Exploratory Analysis of World Bank Population Data Reveals Structural Features

The Day 3 script for Application 5 reads the cleaned World Bank population dataset from Day 2 and performs:

1. Row count and column inventory
2. Summary statistics for the ‘Population‘ field
3. Identification of the top 5 entries by population
4. Missing value check across all columns

The analysis reveals that the dataset includes both individual countries and aggregate regional groupings (e.g., "World", "Low & middle income"). These aggregates must be filtered out before any country-level comparison, illustrating how **source-specific structural features** affect downstream analysis.

GBIF Demonstrates the Full Extract-Clean-Explore Pipeline on a New Source

Application 6 introduces the Global Biodiversity Information Facility (GBIF) as a new source, demonstrating the complete pipeline in a single Day 3 script:

1. **Extract**: fetch 100 occurrence records for *Panthera leo* (lion) via the GBIF API
2. **Clean**: drop records without a year, standardize country names
3. **Explore**: count occurrences by country and record type

Results show that 100% of records are classified as 'HUMAN_OBSERVATION', and that Tanzania, South Africa, and Kenya account for the majority of occurrences — consistent with known lion range distributions. This is a simple but effective **analytic validation** check.

The App's Explore Module Provides No-Code Summaries and Simple Visualizations

The Streamlit Explore module provides:

Feature	Description
Row and column count	Dataset dimensions at a glance
Column type summary	Identifies numeric, categorical, and date fields
Missing value report	Count and percentage of missing values per column
Value counts	Frequency table for categorical fields
Distribution plot	Histogram or bar chart for selected fields
Top-N table	Sortable table of top records by a selected field

These features are sufficient for preliminary plausibility checks and for generating the kind of summary that would appear in a data section of a research paper.

Exploratory Output Should Feed Back into Research Question Refinement

A common pattern in web-based data collection is that exploration **revises the research question**. Participants may discover that:

- The source covers fewer entities than expected
- A key variable has too many missing values to support the intended analysis
- The unit of analysis is more ambiguous than initially assumed
- A more informative variable is available in a related source

This iterative dynamic — question $\rightarrow$ collection $\rightarrow$ exploration $\rightarrow$ revised question — is a normal feature of research design with web-derived data, not a failure.

Participants May Enter the Pipeline at Different Points Depending on Their Situation

Participant Situation	Recommended Entry Point
Has a list of URLs and wants to collect	Day 1 extraction module
Has raw scraped output from another tool	Day 2 cleaning module
Has a partially cleaned CSV and wants to validate	Day 2 validation module
Has a clean dataset and wants to explore	Day 3 explore module
Has a research question but no data yet	Day 1 source identification

The Streamlit app's modular design allows participants to enter at any point without completing all prior steps.

Participants Can Apply Existing Templates to Their Own URLs

The Streamlit app provides source templates for the five application types covered in Days 1 and 2. Participants can:

1. Enter their own URL(s) in the collection module
2. Select the template that most closely matches the source structure (listing, profile, directory, table, API)
3. Preview the extraction result
4. Adjust field selectors if the default template does not match

The template system is designed to lower the barrier to entry for participants who do not have programming experience, while remaining transparent about the underlying extraction logic.

Participants with Existing Data Can Use the App for Cleaning and Exploration Only

Participants who already have a dataset — whether from a previous scraping project, a downloaded CSV, or an institutional database — can upload it directly to the Streamlit app and use only the cleaning and exploration modules.

This use case is important because it demonstrates that the **workflow is modular**: the cleaning and validation logic is useful independently of the extraction logic. A participant who collected data manually from a website can still benefit from the standardized cleaning and documentation tools.

Preliminary Output Should Be Assessed Against Four Criteria Before Use

Before using a web-derived dataset in analysis, participants should assess it against four criteria:

1. **Coverage**: does the dataset represent the intended population, or only a subset?
2. **Consistency**: are fields populated consistently across records, or are there systematic gaps?
3. **Plausibility**: do the values make sense given domain knowledge?
4. **Comparability**: if the dataset will be combined with other sources, are units and field definitions compatible?

These criteria correspond directly to the five validation dimensions introduced on Day 2.

Ethical Considerations Are Not Optional — They Are Part of Research Design

The ethics of web scraping and digital data collection involve at least four distinct concerns:

1. **Legal compliance**: terms of service, copyright, GDPR, CFAA (Computer Fraud and Abuse Act)
2. **Privacy**: scraping personal information without consent, even from public pages
3. **Server load**: aggressive scraping can harm the infrastructure of the source
4. **Representation**: publishing findings based on web-derived data without acknowledging its limitations

These concerns are not merely procedural. They affect the **legitimacy and reproducibility** of research findings.

Terms of Service Vary Widely and Must Be Reviewed Before Collection

Most websites include terms of service that govern automated access. Key considerations include:

- Does the site explicitly prohibit scraping or automated access?
- Does the site provide an official API that should be used instead?
- Is the content protected by copyright, or is it in the public domain?
- Does the site's ‘robots.txt‘ file restrict automated access?

The seminar applications use **official public APIs** (ClinicalTrials.gov, WHO GHO, NIH RePORTER, GovInfo, World Bank, GBIF) — all of which explicitly permit programmatic access. This is the recommended approach for research.

Publicly Accessible Information Is Not Necessarily Appropriate to Collect and Publish

A common misconception is that publicly accessible information is freely available for any research purpose. In practice:

- Personal information (names, contact details, health information) may be protected under GDPR or HIPAA even if publicly posted
- Aggregating individually innocuous data points can create privacy-invasive profiles
- Researchers have an obligation to consider whether collection and publication could harm individuals or communities

The seminar applications focus on **institutional and aggregate data** (trials, grants, bills, countries, species) rather than personal information — a deliberate design choice.

Responsible Scraping Minimizes Impact on Source Infrastructure

Aggressive scraping — making thousands of requests per minute — can harm the infrastructure of the source and may trigger rate limiting or IP blocking. Responsible practices include:

- Using official APIs rather than HTML scraping when available
- Implementing delays between requests (‘time.sleep()‘ in Python, ‘Sys.sleep()‘ in R)
- Caching results locally to avoid repeated requests for the same data
- Respecting ‘Retry-After‘ headers and rate limit responses

All seminar scripts implement local caching and use official APIs, which inherently limits server load.

Published Findings Must Acknowledge the Limitations of Web-Derived Data

When publishing research based on web-derived data, researchers should:

- Describe the source, collection method, and collection date in the methods section
- Report the raw and cleaned record counts
- Acknowledge known coverage gaps and biases
- Provide the collection scripts or a detailed description of the extraction logic
- Archive the raw data (or explain why archiving was not possible)

These practices support replication and allow readers to assess the validity of the findings independently.

Web Scraping Is One Stage in a Multi-Stage Research Workflow

Web scraping is not a self-contained method. It is one stage in a broader research workflow that includes:

1. **Research design**: question formulation, source selection, unit definition
2. **Data collection**: extraction, caching, documentation
3. **Data preparation**: cleaning, standardization, validation
4. **Analysis**: descriptive, inferential, or computational
5. **Interpretation**: findings in context, limitations, implications
6. **Dissemination**: reporting, archiving, replication materials

The seminar covers stages 1–3 in depth and provides a foundation for stages 4–6.

No-Code Tools Lower Barriers but Introduce Their Own Constraints

The no-code Streamlit app is designed to make the workflow accessible to participants without programming experience. However, it has limitations:

- It supports a **fixed set of source templates** — novel or unusual sources may not fit
- It cannot handle **JavaScript-rendered pages** without additional tools
- It does not support **large-scale collection** (thousands of pages)
- It provides **limited customization** of field extraction logic

For participants who need more flexibility, the Python and R scripts in the GitHub repository provide a starting point for extending the workflow beyond the app's capabilities.

APIs Are Preferable to HTML Scraping When Available

Criterion	**API**	**HTML Scraping**
Stability	High — structured, versioned	Low — changes with page redesign
Legality	Explicitly permitted	Depends on terms of service
Data quality	Structured, typed	Requires parsing and cleaning
Rate limits	Documented	Implicit, variable
Coverage	May be incomplete	May capture more context

The seminar applications use APIs exclusively. HTML scraping is introduced conceptually but not demonstrated, because the API-based approach is more robust and more appropriate for research.

Web Pages Also Contain Unstructured Text That Can Be Analyzed Separately

The seminar focuses on **structured extraction** — tables, lists, and API responses that map cleanly to rows and columns. However, many web pages also contain unstructured text (abstracts, descriptions, news articles) that can be analyzed using text-as-data methods.
Possible extensions include:

- Extracting and analyzing grant abstracts from NIH RePORTER
- Analyzing the text of legislative bills from GovInfo
- Applying topic modeling or keyword extraction to clinical trial descriptions

These extensions require additional tools (e.g., ‘spaCy‘, ‘nltk‘, ‘quanteda‘) and are beyond the scope of this seminar, but the structured datasets produced here are natural starting points.

Repeated Collection Over Time Enables Longitudinal Research Designs

A single collection produces a cross-sectional snapshot. Longitudinal research designs require **repeated collection** at multiple time points. Key considerations include:

- Scheduling regular collection runs (e.g., weekly or monthly)
- Tracking changes in records across time points (additions, deletions, modifications)
- Handling schema changes if the source redesigns its structure
- Maintaining a consistent unit identifier across time points

The caching strategy used in this seminar (timestamped raw files) is a simple foundation for longitudinal collection.

Combining Multiple Sources Requires Careful Attention to Unit Compatibility

Many research questions require data from multiple sources. Multi-source integration raises several challenges:

- **Unit compatibility**: do the rows in both datasets refer to the same entities?
- **Temporal alignment**: are the data from the same time period?
- **Field compatibility**: are the same concepts measured in the same way?
- **Coverage overlap**: what proportion of entities appear in both sources?

For example, combining NIH grant data with publication records from PubMed requires a reliable identifier (e.g., grant number) that appears in both sources.

Hour 2 Provides Structured Time for Participants to Work with Their Own Materials

In the second hour of Day 3, participants work independently or in small groups with their own URLs or datasets. The instructor circulates to provide guidance. Participants are encouraged to:

1. Identify one source relevant to their own research
2. Attempt to collect a small sample using the Streamlit app
3. Apply at least one cleaning operation
4. Generate a summary table or value count
5. Note one limitation of the source or the extraction

This structured open practice is the most direct preparation for applying the workflow after the seminar.

Anticipating Common Questions Helps Participants Apply the Workflow Independently

Question	Response
"My source requires login — can I still scrape it?"	Login-protected content raises legal and ethical concerns; use official APIs or data requests instead
"The page structure changes every time I load it"	Dynamic rendering requires browser automation (Selenium, Playwright); beyond this seminar's scope
"My data is in a PDF, not a webpage"	PDF extraction is a separate workflow; tools like 'pdfplumber' or 'tabula' can help
"I only need 20 records — is this overkill?"	Even small collections benefit from documentation and caching for reproducibility
"Can I use this for social media data?"	Platform APIs have strict terms; academic access programs (e.g., Twitter Academic API) are the appropriate route

A Research-Quality Web-Derived Dataset Has Six Identifiable Properties

A web-derived dataset is suitable for research use when it satisfies six criteria:

1. **Source transparency**: the source URL and collection date are documented
2. **Unit clarity**: what one row represents is unambiguous
3. **Field consistency**: all records have the same fields, populated consistently
4. **Cleaning documentation**: all transformations are recorded in a processing log
5. **Known limitations**: coverage gaps and biases are acknowledged
6. **Replicability**: the collection and cleaning scripts are available for inspection

These criteria align with the broader standards for research data management and open science.

Web-Based Data Collection Can and Should Be Integrated into Open Science Workflows

The seminar's emphasis on transparency, documentation, and reproducibility connects directly to open science practices:

- **Open data**: sharing cleaned datasets in public repositories (OSF, Zenodo, GitHub)
- **Open code**: sharing collection and cleaning scripts alongside the data
- **Pre-registration**: documenting the collection strategy before analysis
- **Replication**: enabling other researchers to re-run the collection and verify results

The GitHub repository structure used in this seminar is a direct implementation of these principles.

By End of Day 3, Participants Have Applied the Workflow to Their Own Research Context

By the end of Day 3, participants should be able to:

- Perform basic exploratory analysis on a cleaned web-derived dataset
- Adapt the Streamlit app to their own URLs, URL lists, or existing datasets
- Identify the ethical and legal considerations relevant to their specific collection scenario
- Recognize the limitations of the no-code approach and know when to extend it
- Describe their planned collection workflow in terms of source, unit, fields, and validation strategy

Web-Based Data Collection Is a Transferable Research Competency, Not a One-Off Skill

The three-day seminar has covered a complete workflow from source identification to preliminary analysis. The key takeaway is that this workflow is **transferable**: the same logic applies to any structured online source, in any domain, at any scale.

The specific tools — Streamlit, Python, R, the GitHub repository — are implementations of a more general methodology. Participants who understand the methodology can adapt the tools, replace them, or extend them as their research needs evolve.

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Questions and Discussion

Thank you!

Questions?

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